

MultiTech Escape PUZZLE GAME

A complete Unity game template for creating arrow rope puzzle games with customizable themes and level management.

Overview

MultiTech Escape PUZZLE GAME is a Unity game template that provides a complete foundation for building arrow rope puzzle games. The template includes:

- **Level Editor:** Visual level creation and editing tools
- **Theme System:** Customizable themes with multiple color options
- **Grid Management:** Flexible grid-based gameplay system
- **Arrow System:** Advanced arrow movement and collision detection
- **UI System:** Complete UI management for menus, gameplay, and level selection

Quick Start

Installation

1. Import the package into your Unity project (Unity 2021.3 or later recommended)
2. Open the main scene: `Assets/MultiTech Studio/Escape Game/Scenes/Level Screen.unity`
3. Access the setup window: `Tools > MultiTech Studio > Escape Game Setup`

First Steps

1. Setup Themes:

- Open `Tools > MultiTech Studio > Escape Game Setup`
- Navigate to the "Theme" tab
- 7 default themes are automatically created on first launch
- Customize themes by selecting them and editing their properties

- Click "APPLY" to apply a theme to the current scene
- (Optional) Add `ThemeManager` component to enable runtime theme switching

2. Configure Game Settings:

- Adjust game parameters in the LevelManager and GridManager components
- Customize UI elements in the UiManager component
- (Optional) Setup runtime theme system by adding ThemeManager component

Features

Theme System

The theme system allows you to create and manage multiple visual themes for your game:

- **Theme Icon:** Visual representation for theme selection
- **Arrow Prefab:** Custom arrow prefab for each theme
- **Color Options:** Single or multiple colors per theme
- **Skybox:** Custom skybox material per theme
- **Grid Icon:** Custom grid dot sprite per theme
- **Background:** Optional background sprite

Editor-Based Theme Application:

- Apply themes directly in the editor using the Theme Editor window
- Themes are applied to scene objects and prefabs
- Automatically creates `CurrentTheme.asset` for runtime access

Runtime Theme System (Optional):

- **CurrentTheme ScriptableObject:** Holds the reference to the active theme
- **ThemeManager Component:** Applies themes at runtime and supports theme switching
- Add `ThemeManager` to your scene for runtime theme support
- Themes can be switched during gameplay
- Fully backward compatible - works without runtime system

Note: If a selected sprite appears too small or too big when applied via the editor window, adjust its **Pixels Per Unit** setting in the sprite's Import Settings. Select the sprite in the Project window, then in the Inspector, adjust the "Pixels Per Unit" value (typically between 100-1000 depending on your sprite resolution).

SDK Integration

The template includes optional integration with Google Mobile Ads (AdMob) and Firebase:

- **AdMob Integration:** Full AdMob support with banner, interstitial, and rewarded ads
- **Firebase Integration:** Firebase analytics and crash reporting
- **Conditional Compilation:** Safe compilation with or without SDKs installed
- **Auto-Detection:** Automatic SDK detection and define management
- **Easy Configuration:** Configure SDKs through the Settings tab in the setup window

Access SDK Settings:

1. Open `Tools > MultiTech Studio > Escape Game Setup`
2. Click on the "Settings" tab
3. View SDK status, enable/disable SDKs, and configure AdMob settings

AdMob Configuration:

- Add `AdsManager` component to a GameObject in your scene
- Configure ad unit IDs in the Settings tab
- Set interstitial level threshold and display preferences
- All AdMob code is conditionally compiled - project compiles safely without AdMob

Firebase Setup:

- Add `FirebaseManager` component to a GameObject in your scene
- Configure Firebase in your Firebase console
- Automatic initialization when enabled
- All Firebase code is conditionally compiled - project compiles safely without Firebase

Level Editor

- Visual grid-based level creation

- Arrow path painting tools
- Real-time preview
- Level asset serialization
- Easy level management

Gameplay Features

- Smooth arrow movement with DOTween animations
- Collision detection between arrows
- Exit detection when arrows leave the grid
- Level progression system
- Coin and lives management
- Runtime theme switching (optional)

Project Structure

```
Assets/MultiTech Studio/Escape Game/
├── Animations/           # Animation controllers and clips
├── Documents/           # License and documentation
├── Editor/              # Editor scripts and tools
│   ├── EscapeGameSetupWindow.cs
│   ├── LevelSaveTool.cs
│   └── ArrowLineBrushEditor.cs
├── Fonts/               # Font assets
├── Levels/              # Level ScriptableObject assets
├── Materials/           # Material assets
├── Prefabs/             # Game prefabs (Arrows, Snakes, Trains, etc.)
├── Scenes/              # Unity scenes
├── Scripts/             # Runtime scripts
│   ├── ThemeData.cs     # Theme configuration ScriptableObject
│   ├── CurrentTheme.cs  # Active theme reference ScriptableObject
│   ├── ThemeManager.cs  # Runtime theme manager component
│   ├── LevelAsset.cs    # Level data ScriptableObject
│   ├── ArrowLine.cs     # Arrow line behavior
│   ├── GridManager.cs   # Grid management
│   ├── LevelManager.cs  # Level loading and management
│   ├── AdsManager.cs    # AdMob integration (optional)
│   └── FirebaseManager.cs # Firebase integration (optional)
```

```
|   └─ ...
|   └─ Editor/                               # Editor scripts
|       └─ EscapeGameSetupWindow.cs        # Main setup window
|       └─ ThemeOperations.cs               # Theme apply/delete operations
|       └─ ThemeHelpers.cs                 # Theme helper functions
|       └─ AutoDefines.cs                  # SDK auto-detection
|       └─ ...
|   └─ Sprites/                             # Sprite assets
|   └─ Themes/                              # Theme ScriptableObject assets
|       └─ CurrentTheme.asset              # Active theme reference (auto-created)
|       └─ Theme_*.asset                   # Theme configuration assets
```

Requirements

- Unity 2021.3 LTS or later
- Universal Render Pipeline (URP)
- DOTween (included in package)

Optional Dependencies

- **Google Mobile Ads SDK:** For AdMob integration (optional)
- **Firebase SDK:** For Firebase analytics and crash reporting (optional)

Note: The template works perfectly without these SDKs. They are optional and can be enabled/disabled through the Settings tab.

Support

For detailed documentation, see [Document.md](#).

For issues or questions, please refer to the project documentation or contact support.

License

See [Assets/MultiTech Studio/Escape Game/Documents/LICENSE.txt](#) for license information.